

Class Incremental Learning for IoT Intrusion Detection: A Transformer-Based Approach

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Abstract—Network Intrusion Detection Systems (NIDS) for IoT environments must continuously adapt to new attack types without forgetting previous ones — a problem known as catastrophic forgetting in the context of Class Incremental Learning (CIL). In this work we compare standard CIL approaches (BiC, Memento) on CNN and Transformer architectures, and propose an adaptation of ROBUSTA — originally designed for Few-Shot CIL on images — to the IoT network traffic domain. Results show that, given the same standard CIL approach, CNN and Transformer offer comparable performance. However, leveraging Transformer-native mechanisms — specifically prefix tuning through delta parameters — produces significant advantages in more challenging scenarios: in the sequential scenario (+1+1+1+1+1+1), ROBUSTA outperforms BiC and Memento with a margin of about 38 percentage points on plasticity (C^{new}), while reducing forgetting on C^{old} by 8–12 percentage points. In the few-shot scenario with only 153 available samples, ROBUSTA achieves the best F1 New (97.4%), confirming its design for data-scarce scenarios. In single-increment scenarios (+1, +6), standard approaches remain competitive, suggesting that the complexity of the incremental scenario is the determining factor in choosing the optimal CIL strategy.

Index Terms—Class Incremental Learning, Network Intrusion Detection, IoT Security, Catastrophic Forgetting, Transformer, Prefix Tuning, Few-Shot Learning, Bias Correction, Knowledge Distillation

I. INTRODUCTION

The growing diffusion of IoT devices has significantly expanded the attack surface of modern networks, making Network Intrusion Detection Systems (NIDS) a critical component for the security of IoT infrastructures [5]. Unlike traditional environments, IoT systems are characterized by heterogeneous traffic, limited computational resources, and the continuous emergence of new attack types — often with very few samples available for training. These characteristics make static detection approaches inadequate, as they require costly complete re-training every time a new threat is identified.

Class Incremental Learning (CIL) presents itself as a natural paradigm to address this problem, allowing models to learn new classes incrementally without accessing historical data. However, the main obstacle of CIL is *catastrophic forgetting* [6] — the tendency of models to degrade performance on previous classes when updated with new data. Existing mitigation strategies, such as Bias Correction (BiC) [3] and Memento [7], combine rehearsal-based learning and knowledge

distillation, but have been designed and evaluated primarily on convolutional architectures (CNN) for image classification.

Recently, Transformer models [8] have shown promising capabilities also in the domain of network traffic classification, thanks to the self-attention mechanism that captures global dependencies in packet sequences. However, it remains an open question whether Transformer models offer intrinsic advantages over CNNs in CIL scenarios, and above all whether CIL approaches specifically designed to leverage the native properties of the Transformer architecture exist.

In this work we address these questions through a systematic experimental evaluation on TON-IoT [9], a realistic IoT traffic dataset with 12 attack classes. We compare BiC and Memento on CNN and Transformer architectures across three incremental scenarios (+1, +1+1+1+1+1+1, +6), and propose an adaptation of ROBUSTA [4] — originally designed for Few-Shot CIL on images — to the IoT network traffic domain. The main contributions of this work are:

- A systematic comparison between CNN and Transformer architectures with standard CIL approaches, showing comparable performance given the same method.
- The adaptation of ROBUSTA to the IoT domain, with original modifications to the PredictionNet loss, knowledge distillation, replay balancing, and task inference.
- The demonstration that Transformer-native approaches produce significant advantages in multi-step scenarios: in the 1+1+1+1+1+1 scenario, ROBUSTA outperforms BiC and Memento with a margin of about 38 percentage points on plasticity (C^{new}) and reduces forgetting on C^{old} by 8–12 percentage points.
- The analysis of the impact of the number of incremental steps and the number of base classes, with evidence that scenario complexity is the determining factor in choosing the optimal CIL strategy.

The rest of the paper is organized as follows: Section II discusses related work, Section III describes the methods, Section IV the experimental setup, Section V presents the results, and Section VI concludes the work.

II. RELATED WORK

CIL approaches applied to NIDS can be divided into three main families [1]: fine-tuning, fixed-representation, and model-growth. Among forgetting mitigation strategies, the